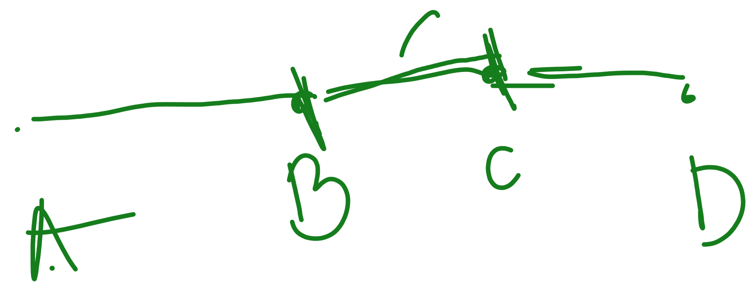
All Optimizers

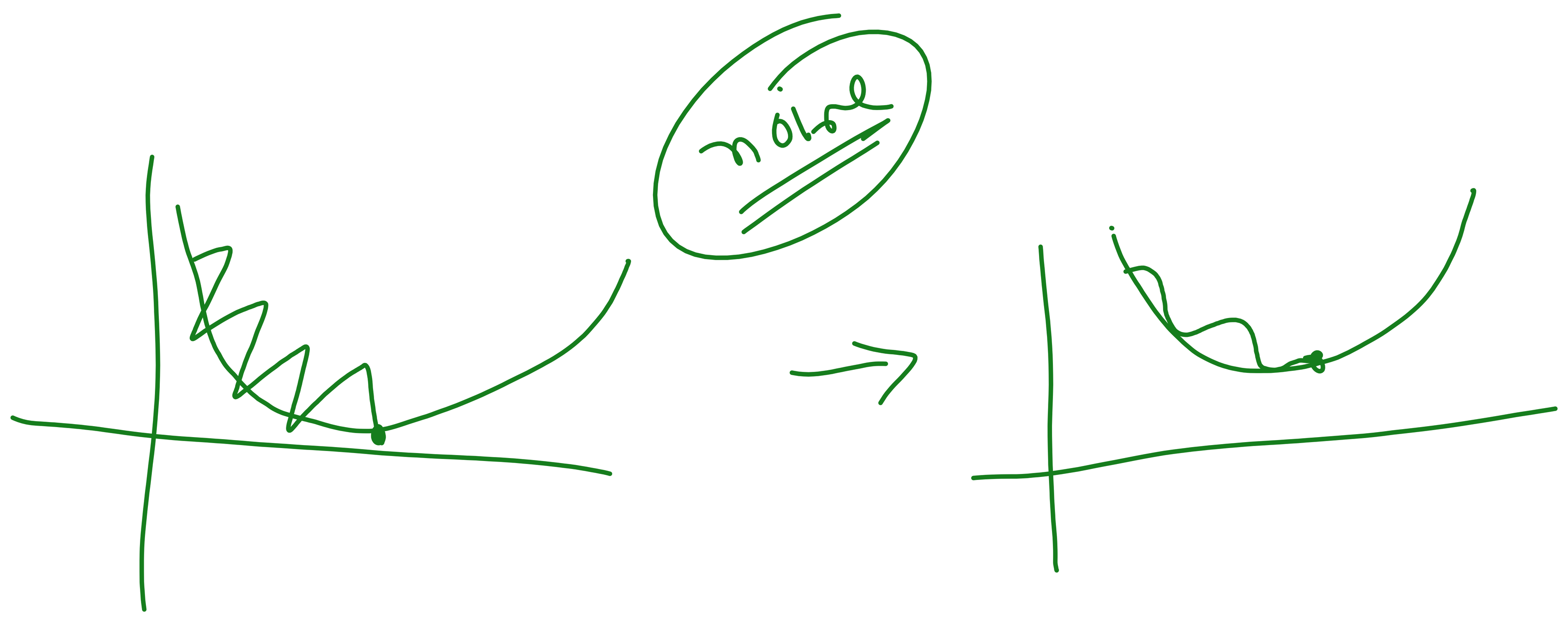
[redacted]

SGD with momentum:

1. momentum → Smoothing
2. dynamic learning rate:



Objective:



EWMA

weight updation formula

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$$

$$w_{t+1} = w_t - \boxed{\eta \frac{\partial L}{\partial w_t}}$$

→ weight updation with momentum:

$V_{dw} = 0$

$$V_{dw_t} = \beta \underbrace{(V_{dw_{t-1}})}_{} + (1-\beta)\frac{\partial L}{\partial w_{t-1}}$$

0.75 75%   0.25 25%

$$w_t = w_{t-1} - \eta V_{dw_{t-1}}$$

Adaptive gradient
Adagrad

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$$

η

error

$$\eta' = \frac{}{\sqrt{\alpha_t + \epsilon}}$$

no momentum
dynamic η

$$w_{new} = w_{old} - \frac{\eta}{\sqrt{\alpha_t + \epsilon}} \cdot \frac{\partial L}{\partial w_{old}}$$

$\alpha_t \rightarrow$ avg σ

$$\alpha_t = \sum_{i=1}^{t} \left(\frac{\partial L}{\partial w_t} \right)^2$$

Core Problem of AdaGrad

👉 **The learning rate keeps shrinking continuously and eventually becomes too small**

RMS prop

$$\eta' = \frac{\eta}{\sqrt{S_{dw} + \epsilon}}$$

$$E.W.A \rightarrow \left(\frac{\partial L}{\partial w_t} \right)^2$$

$$S_{dw_t} = \beta S_{dw_{(t-1)}} + (1-\beta) \left(\frac{\partial L}{\partial w_t} \right)^2$$

Adam

$$w_{t+1} = w_t - \frac{\eta}{\sqrt{S_{dw_t} + \epsilon}} * V_{dw_t}$$

→ weight update

$$b_{t+1} = b_t - \frac{\eta}{\sqrt{S_{db_t} + \epsilon}} * V_{db_t}$$